THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET B

Chapter 7: **Particulate Nature of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The spaces between the particles of matter are called: 1

(a) vacuoles **(b)** cells
(c) pores of skin **(d)** interparticle spaces

2. Interparticle space is maximum in: 1

(a) solids **(b)** ice at 0 °C only **(c)** gases **(d)** liquids

3. The particles in a solid: 1

(a) move in straight lines only **(b)** move freely everywhere
(c) are absent **(d)** vibrate about fixed positions

4. Compressing a gas makes its particles: 1

(a) come closer **(b)** stop moving
(c) disappear **(d)** move further apart

5. The melting point of iron is about: 1

(a) 133 °C **(b)** 1538 °C **(c)** 0 °C **(d)** 100 °C

6. During melting, the interparticle forces of attraction: 1

(a) turn infinite **(b)** remain unchanged **(c)** become stronger **(d)** become weaker

7. A drop of ink spreads in water because: 1

(a) ink melts **(b)** water freezes
(c) particles of ink and water mix **(d)** water evaporates

8. Which statement about ice is correct? 1

(a) It is more dense than water
(b) It has no particles
(c) Its particles are farther apart than those in water
(d) It is a gas

9. If all particles are removed from a chair, then: 1

(a) nothing changes **(b)** it turns into a gas
(c) it weighs less but remains **(d)** nothing of the chair remains

10. A gas takes the volume of its container because: 1

(a) its particles are fixed
(b) its particles move freely in all directions
(c) it has no particles
(d) it has strong forces

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why do gases have no fixed shape and volume? 2

12. Define melting point. Give the value for ice. 2

13. Define boiling point. 2

14. Why does a gas get compressed easily but water does not? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Why do gases mix easily but solids do not? 3

16. Give two reasons why solids have fixed shape. 3

17. Explain why liquids have definite volume but take the shape of the container. 3

OR

Rice flour and grains take the shape of the jar. Are they liquids? Explain. 3

18. Draw the arrangement of particles in aluminium foil, glycerin and methane gas. 3

OR

Explain why the water in the ocean tastes salty even though salt is not seen. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

19. Describe with examples how the diffusion of ink, perfume and sugar shows that particles move. 5

OR

Explain the table of melting points: ice 0 °C, urea 133 °C, iron 1538 °C, and relate to particle forces. 5

20. Why are solids rigid, liquids flowing and gases spreading? Answer using forces and spaces. 5

OR

Describe the three states of water with particle diagrams and changes between them. 5

END OF QUESTION PAPER Total: 40 Marks

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